



CEO aging and investment propensity: Experimental evidence

Steffen Brenner^{a,*}, Jeppe Christoffersen^b, Torben Andersen^a, Thomas Plenborg^b

^a Copenhagen Business School, Department of International Economics, Government, and Business, Porcelænshaven 24A, 2000 Frederiksberg, Denmark

^b Copenhagen Business School, Department of Accounting, Solbjergs Plads 3, 2000 Frederiksberg, Denmark

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ABSTRACT

Firms led by older CEOs invest less, but the mechanism behind this age–investment gradient remains unclear. We study how CEO age shapes investment evaluation by eliciting CEOs' ratings of a project's investment attractiveness in a large-scale scenario-based experiment with more than 3700 CEOs from owner-managed Danish firms. Our design provides direct evidence on CEOs' investment evaluations by separating baseline differences in investment attractiveness from sensitivity to risk moments and payout horizon. Each CEO evaluates randomized capital-budgeting scenarios relative to the status quo, allowing us to distinguish a baseline shift in willingness to view projects as compelling from age differences in responsiveness to risk characteristics and payout horizon. We document a pronounced negative association between CEO age and investment attractiveness. Holding expected IRR, variance, skewness, and payout horizon constant, CEOs aged 60 or older rate the same opportunities as less attractive, while age differences in sensitivity to these attributes are weak. Standard stated risk attitudes and a lottery measure show no systematic age pattern. For external validity, we construct a CEO-specific inaction propensity from the experiment—the residual tendency to rate otherwise identical projects as less attractive after accounting for project characteristics—and show that it predicts lower investment in administrative data. Including this measure attenuates the reduced-form age–investment association, suggesting that age-related investment declines are not readily explained by conventional risk preferences or investment horizons alone, but also reflect a broader residual tendency towards lower investment propensity.

1. Introduction

Corporate investment is a central capital-allocation decision and a primary lever through which CEOs shape firm growth. Yet a consistent empirical regularity is that firms led by older CEOs exhibit lower growth and invest less (Child, 1974; Cline and Yore, 2016; Belenzon et al., 2019). More broadly, older-CEO firms initiate fewer major strategic initiatives—such as acquisitions and R&D investment—and are less likely to pursue international expansions than firms led by younger CEOs (Barker III and Mueller, 2002; Matta and Beamish, 2008; Levi et al., 2010; Yim, 2013; Serfling, 2014; Kang, 2016). Using quasi-experimental variation in CEO age, Belenzon et al. (2019) support a causal interpretation of the CEO-age–investment link rather than simple selection of older CEOs into low-investment firms. As executive populations age—among U.S. public firms, the share of CEOs aged 60+ has risen to about 40%—

* Corresponding author.

E-mail address: sbr.egb@cbs.dk (S. Brenner).

understanding age-related shifts in strategic decision making is increasingly important.¹

Why does CEO age predict corporate investment? Prior work offers several candidate channels. One prominent view emphasizes age-related differences in choices under uncertainty, treating older CEOs as more conservative in corporate policies and risk exposure (Tufano, 1996; Milidonis and Stathopoulos, 2014; Malmendier et al., 2011). A second explanation focuses on horizons, arguing that older CEOs may have weaker incentives to initiate long-gestation projects whose benefits arrive after they expect to leave office. A third, broader account points to age-related influences beyond conventional risk preferences and investment horizons that may reduce investment propensity. For example, reduced energy, lower cognitive stamina, and greater aversion to complexity may make CEOs more selective about undertaking demanding initiatives even when expected returns are favorable. This paper seeks to distinguish between these alternatives. Specifically, we contrast risk-preference and horizon-based accounts with a broader residual explanation in which CEO age may influence investment through multiple mechanisms. Rather than identifying these mechanisms individually, our design tests whether a residual age effect remains after accounting for the benchmark explanations.

Existing evidence struggles to pin down the mechanisms behind the CEO age–investment relationship because realized investment is an indirect proxy for managerial decision making. Lower investment may reflect how CEOs evaluate, implement, and respond to uncertain opportunities rather than a stable increase in risk aversion. Consistent with this concern, available CEO-level evidence shows only weak links between age and standard measures of risk aversion (Brenner, 2015), and recent work cautions against interpreting reduced-form CEO-age effects as evidence of any single underlying mechanism (Trabert, 2023).

Moreover, older CEOs appear not only to initiate fewer expansions but also to engage less in strategic reallocation—implementing fewer strategic changes, restructuring more slowly, and divesting underperforming units less frequently (Grimm and Smith, 1991; Wiersema and Bantel, 1992; Huber et al., 1993; Datta et al., 2003; Zhang and Rajagopalan, 2010; Li et al., 2017; Kalasin, 2023). Because divestments and restructurings often reduce risk exposure, the combination of lower investment and lower divestment is difficult to reconcile with conventional risk aversion alone and instead suggests that broader mechanisms may influence CEOs' propensity to initiate strategic action. Finally, horizon accounts imply that age differences should be strongest for long-gestation projects and should respond to exogenous variation in payout timing—predictions that have rarely been tested directly (e.g., Brickley et al., 1999; Cline and Yore, 2016).

Investment decisions are not only about evaluating return distributions; they also require initiating and implementing organizational action. Consequently, age-related differences in investment may arise through several mechanisms beyond conventional risk preferences and investment horizons. For example, major investments require information acquisition, organizational coordination, and sustained execution, implying that higher perceived implementation costs or cognitive effort may reduce willingness to act (Cyert and March, 1963; March and Simon, 1958; Bertrand and Mullainathan, 2003; Borgschulte et al., 2025; Hess, 2014; Queen et al., 2012). In addition, older CEOs may hold different beliefs about investment opportunities, pursue different managerial objectives or legacy concerns, or evaluate opportunities differently because of accumulated managerial experience (Hambrick and Mason, 1984; March and Shapira, 1987; Brickley et al., 1999). When “no investment” represents an explicit default, certainty weighting and status quo bias may further increase the attractiveness of maintaining the current course of action (Samuelson and Zeckhauser, 1988; Kahneman and Tversky, 1979; Prelec, 1998; Wakker, 2010). Although conceptually distinct, these mechanisms share a common behavioral implication: they lower investment propensity. Accordingly, holding expected IRR, variance, skewness, and payout horizon constant, they imply a baseline downward shift in Investment Attractiveness rather than systematically stronger responsiveness to project risk moments or payout timing.

We conduct a large-scale, scenario-based experiment with approximately 3700 CEOs from owner-managed Danish firms to separate baseline willingness to invest from sensitivity to risk moments and payout horizon. We operationalize “older” CEOs as those aged 60 and above, consistent with prior evidence of nonlinear age patterns in investment-related corporate policies (Yim, 2013; Belenzon et al., 2019) and with psychology research documenting affective and cognitive shifts in late midlife (Carstensen, 2006; Mather and Carstensen, 2005). To ensure our findings do not hinge on this threshold, we conduct sensitivity tests using alternative cutoffs. Each respondent evaluates a realistic investment opportunity presented in a familiar scenario-analysis format (worst/base/best), enabling us to observe CEOs' ratings of the project's investment attractiveness, that is, how compelling the project is relative to maintaining the status quo. Key project attributes are orthogonally randomized. From the randomized worst/base/best outcomes in each table, we calculate the implied return distribution's first three moments—expected IRR, variance, and skewness—for use in the empirical analysis, and we independently randomize payout horizon. This design yields causal estimates of how CEOs map return levels, risk moments, and timing into investment attractiveness. It also allows us to test whether a residual age-related difference in investment attractiveness remains after accounting for these experimentally manipulated project characteristics. To evaluate competing explanations, we link respondents to Danish administrative records (e.g., education, wealth, tenure, firm characteristics) and elicit risk preferences using both self-reported risk attitudes and a Holt and Laury (2002) lottery choice task.

Our results reveal a substantial residual age-related difference in investment attractiveness beyond conventional risk preferences and investment horizons. Holding expected IRR, variance, skewness, and payout horizon constant, CEOs aged 60 or older rate the same investment opportunity as less attractive. By contrast, age differences in responsiveness to the randomized risk moments and payout horizon are weak, and independent measures of risk preferences (stated risk attitudes and an incentivized lottery task) show no systematic age pattern. These findings are robust to controls for CEO and firm characteristics and suggest that the age–investment gradient reflects a broader residual explanation involving multiple mechanisms.

¹ Own calculations are based on data of listed firms in financial years 2014 and 2024 from ExecuComp.

The experimental measure also predicts real investment. We construct a CEO-specific inaction propensity from the experiment—the residual tendency to rate otherwise identical projects as less attractive after accounting for project characteristics, measured risk preferences, and investment horizons—and link it to Danish administrative data. CEOs with higher inaction propensity lead firms that invest less in the field. Moreover, including this measure in standard investment regressions attenuates the reduced-form association between CEO age and realized investment while leaving it economically meaningful, consistent with part of the age–investment relationship operating through the residual behavioral component captured by our experiment.

We contribute to the corporate finance literature on CEO effects and corporate policies in three ways. First, we provide direct evidence on how CEO age shapes investment decisions at the point when opportunities are evaluated. Whereas prior work largely infers mechanisms from realized corporate outcomes, our approach complements the “managerial styles” tradition in corporate finance (Bertrand and Schoar, 2003; Graham et al., 2013) by directly observing how CEOs assess otherwise identical investment opportunities. This allows us to distinguish systematic age differences in investment evaluation from differences inferred solely from realized corporate policies.

Second, we refine how CEO age should be interpreted. The common view equates older CEOs' lower investment with higher risk aversion, but our evidence suggests that this interpretation is incomplete. After accounting for conventional risk preferences and investment horizons, we document a substantial residual age-related difference in investment attractiveness. We interpret this residual as reflecting a broader tendency towards lower investment propensity that may arise through several age-related influences—including implementation frictions, greater cognitive effort, stronger status quo tendencies, accumulated managerial experience, changing managerial objectives, and heightened certainty weighting. By also eliciting both survey-based and lottery-based measures of risk preferences (Dohmen et al., 2011; Holt and Laury, 2002), we show that this residual behavioral component is distinct from conventional proxies for risk and loss aversion.

Third, we bridge experimental measurement and archival outcomes, contributing to the behavioral corporate finance literature linking managerial traits and beliefs to corporate policies (Malmendier and Tate, 2005; Malmendier and Tate, 2008; Malmendier et al., 2011). We construct a CEO-specific inaction propensity from the experiment and link it to Danish administrative records. This experimentally derived measure predicts firms' realized investment behavior and attenuates the reduced-form association between CEO age and investment. More broadly, our findings suggest that CEO age is associated with a residual dimension of managerial decision making beyond conventional risk preferences and investment horizons that helps explain realized corporate investment.

2. Theory and hypotheses

2.1. CEO age and the investment decision

A large corporate finance literature shows that individual CEOs shape corporate policies and outcomes (Bertrand and Schoar, 2003; Graham et al., 2013). A robust empirical regularity is that firms led by older CEOs exhibit lower economic activity—most prominently, lower growth and investment (Child, 1974; Cline and Yore, 2016; Belenzon et al., 2019). Consistent with this pattern, older-CEO firms undertake fewer major expansions, including acquisitions, innovation investments, CSR initiatives, and international growth initiatives (Barker III and Mueller, 2002; Matta and Beamish, 2008; Levi et al., 2010; Yim, 2013; Serfling, 2014; Kang, 2016). Importantly, Belenzon et al. (2019) provide quasi-experimental evidence consistent with a causal CEO-age–investment link rather than selection of older CEOs into low-investment firms.

Corporate investment can be conceptualized as an action choice. Initiating a project requires commitment, coordination, and follow-through, whereas inaction preserves the status quo. Prior research in strategy and organizational theory emphasizes that strategic change involves mobilizing organizational resources and overcoming internal frictions, while maintaining existing policies typically requires less managerial intervention (e.g., Cyert and March, 1963; Hannan and Freeman, 1984). Investment and divestment decisions are therefore central levers through which CEOs reshape firm policy and exposure. They also carry personal stakes—success can enhance reputation and compensation, while failure erodes wealth and status (Brickley et al., 1999; Edmans et al., 2017).

Importantly, the empirical pattern documented in prior work is not limited to lower investment rates among older CEOs. Older CEOs not only initiate fewer investments and expansions; they also implement fewer strategic changes, restructure more slowly, and divest underperforming units less frequently (Grimm and Smith, 1991; Wiersema and Bantel, 1992; Huber et al., 1993; Datta et al., 2003; Zhang and Rajagopalan, 2010; Li et al., 2017; Kalasin, 2023). The combination of lower investment and lower divestment is difficult to reconcile with a pure increase in risk aversion, as divestment typically reduces risk exposure. Instead, these patterns suggest that conventional risk preferences alone may not fully account for the age–investment relationship.

2.2. Competing explanations for the CEO age–investment gradient

2.2.1. Risk-preference channel (variance sensitivity and loss aversion)

In corporate finance, investment behavior is typically framed as a tradeoff between expected returns and risk. Because higher expected returns are universally desirable, cross-sectional differences in investment propensity are often attributed to differences in how decision makers perceive and weight risk, rather than to differences in the valuation of returns. This makes a risk-preference channel a natural benchmark explanation for the CEO age–investment gradient.

In expected-utility terms, risk aversion implies a dislike of dispersion: holding expected return constant, a more risk-averse CEO finds projects with higher variance less attractive (Markowitz, 1952; March and Shapira, 1987). In our moment-based design, a variance-driven account therefore predicts that older CEOs should reduce Investment Attractiveness more sharply as IRR dispersion

increases.

A related mechanism is loss aversion—the tendency to overweight downside outcomes relative to equivalent upside gains (Kahneman and Tversky, 1979; Tversky and Kahneman, 1992). Loss aversion implies stronger negative sensitivity to lower-tail outcomes. Importantly, recent work suggests that loss aversion may decline with age, which would predict weaker downside sensitivity among older decision makers (Pachur et al., 2017; Frank and Seaman, 2023).

We use skewness to capture tail asymmetry: positive skewness increases rare upside potential at a given mean and variance. Responses to skewness may reflect greater valuation of rare upside outcomes, weaker weighting of downside outcomes, or nonlinear probability weighting. We therefore interpret skewness effects as a reduced-form indicator of tail weighting rather than as identifying any one of these mechanisms (Kahneman and Tversky, 1979; Tversky and Kahneman, 1992; Prelec, 1998).

Taken together, a risk-preference account implies that age differences should emerge primarily in slope responses to variance and skewness (risk moments).

2.2.2. Horizon channel

A second benchmark explanation emphasizes the timing of payoffs. Many corporate investments require immediate commitments, while benefits arrive with delay and uncertainty. If older CEOs have shorter remaining career horizons—because they expect to retire sooner or place less weight on post-tenure outcomes—they may be less willing to initiate long-gestation projects whose benefits materialize after they anticipate leaving office. This logic aligns with career-concern and incentive-based models in which managers' private returns depend on when outcomes become observable (Brickley et al., 1999; Edmans et al., 2017). Empirically, proximity to retirement has been linked to lower investment in capital expenditure, innovation, acquisitions, and CSR (Matta and Beamish, 2008; Antia et al., 2010; Serfling, 2014).

Horizon considerations need not imply mechanical short-termism. Late-career CEOs may also prioritize legacy and downside protection, becoming more cautious about initiatives that risk reputational or performance setbacks that cannot be repaired before retirement (Hambrick and Mason, 1984; Hambrick and Finkelstein, 1987). In SMEs—where CEOs often have concentrated ownership and broad discretion—such horizon-related motives may translate directly into investment policy.

The key empirical implication is straightforward. If horizon considerations are a primary driver of the CEO age–investment gradient, older CEOs should be more sensitive to payout horizon: holding the return distribution constant, longer payout horizons should reduce Investment Attractiveness more strongly for older CEOs. Conversely, if randomized variation in payout timing does not meaningfully alter the age gradient, horizon-based explanations are less likely to account for the observed pattern. Finally, age should not be viewed simply as a proxy for career horizon. As Antia et al. (2010) argue, age, tenure, and retirement proximity capture distinct influences on managerial behavior, making it important to distinguish horizon-based explanations from other age-related mechanisms.

2.2.3. Investment propensity beyond risk preferences and horizons

A third, broader explanation emphasizes mechanisms that influence CEOs' investment propensity beyond conventional risk preferences and investment horizons. Corporate investments—especially in SMEs—often require CEO time and sustained follow-through, including information gathering, organizational coordination, and execution. Consequently, investment decisions may be shaped not only by expected returns and risk, but also by broader influences on the willingness to initiate organizational action. Although conceptually distinct, these influences share a common behavioral implication: they reduce the investment propensity, even when expected returns are favorable.

Several complementary mechanisms may contribute to lower investment propensity. Aging is associated with changes in cognitive resources, motivation, and self-regulation that can raise the subjective cost of high-effort tasks (Kanfer and Ackerman, 2004; Hess, 2014). Consistent with this possibility, shocks to executive capacity affect firm outcomes (Bennedson et al., 2020), while recent evidence suggests that executive aging is accompanied by changes in stress and executive capacity that may similarly influence managerial decision making (Borgschulte et al., 2025). Managers may also derive utility from avoiding demanding organizational initiatives, consistent with the notion of a managerial “quiet life” (Bertrand & Mullainathan, 2003). Life-cycle models additionally predict greater weight on leisure and preservation motives later in life, which can reduce willingness to undertake costly initiatives (Ando and Modigliani, 1963; Gourinchas and Parker, 2002). At the same time, accumulated managerial experience and changing managerial objectives may influence how otherwise similar investment opportunities are evaluated, independently of conventional risk preferences or investment horizons (Hambrick and Mason, 1984; March and Shapira, 1987). For owner-managed firms, concentrated downside exposure and non-pecuniary objectives may further increase the appeal of preserving the status quo.

The presence of a sure default represents another potential mechanism. When choices juxtapose certainty and uncertainty, decision makers often overweight the sure option (the certainty effect), a pattern driven by nonlinear probability weighting rather than variance sensitivity per se (Kahneman and Tversky, 1979; Prelec, 1998; Wakker, 2010). In our setting, an explicit no-investment (sure) alternative may therefore be disproportionately attractive if older CEOs place greater weight on certainty or perceive greater costs of initiating organizational action.

Although conceptually distinct, the influences discussed above share a common empirical implication. Holding expected IRR, variance, skewness, and payout horizon constant, they predict a baseline downward shift in Investment Attractiveness rather than systematically stronger responsiveness to project risk moments or payout timing. This baseline-shift implication is the prediction we take to the data and formalize in H1 below.

2.3. Hypotheses

Let Investment Attractiveness denote the CEO's evaluation of how compelling a project is relative to maintaining the status quo. For the empirical analysis, we calculate the first three probability-weighted moments of the implied IRR distribution for each vignette: expected IRR, standard deviation of IRR (dispersion), and skewness of IRR (tail asymmetry). We additionally vary payout horizon. All hypotheses are stated holding constant the other listed project attributes.

H1. Baseline investment propensity. Holding expected IRR, standard deviation of IRR, skewness of IRR, and payout horizon constant, CEOs aged 60 or above rate the same project as less attractive, consistent with a lower baseline investment propensity.

H2. Horizon sensitivity. Longer time-to-payout reduces Investment Attractiveness more for CEOs aged 60 or above than for younger CEOs, conditional on expected IRR and the IRR risk moments.

H3. Risk-moment sensitivity. If the age–investment gradient is driven by risk-preference channels, age differences should emerge primarily in sensitivity to the IRR risk moments.

H3a. Variance. Investment Attractiveness decreases more strongly with higher IRR variance for CEOs aged 60 or above.

H3b. Skewness. Investment Attractiveness responds differently to skewness for CEOs aged 60 or above, reflecting differences in tail weighting.

3. Research design

3.1. Data collection

3.1.1. Survey experiment

Our hypotheses refer to constructs—certainty sensitivity, responsiveness to downside risk, and symmetric risk aversion—that are difficult, if not impossible, to observe through archival data. To directly assess these mechanisms, we conducted a large-scale online field survey experiment targeting CEOs of Danish small and medium-sized, owner-managed enterprises.

We selected owner-managed Danish firms as the target population because their CEOs typically hold primary responsibility for investment decisions and are accessible for large-scale data collection. Denmark also offers ideal conditions for linking survey responses to high-quality administrative records, including reliable information on executive demographics, firm characteristics, and financials.

To elicit CEOs' investment preferences under uncertainty, we presented each respondent with a hypothetical investment scenario, structured using a scenario analysis framework widely used in capital budgeting (Healey and Hodgkinson, 2008). The vignette included best-case, base-case, and worst-case outcomes, each defined in terms of projected returns and associated probabilities. Respondents rated the Investment Attractiveness of each project on a 7-point Likert scale (1 = very unattractive, 7 = very attractive). We interpret lower investment-attractiveness ratings as indicating a greater reluctance to pursue the project, which lets us map systematic level differences in ratings to differences in baseline investment propensity even though the task does not require an explicit binary invest/no-invest choice.²

By systematically varying probabilities and returns across scenarios, we can separately estimate (a) baseline investment propensity (H1); (b) responsiveness to longer time horizons (H2); and (c) sensitivity to the first three moments of the return distribution, including variance (H3a) and skewness (H3b).

Table 1 illustrates a sample investment vignette and Appendix A provides an overview of the full vignette design and parameter variation. The vignette parameters were drawn from discrete distributions spanning wide ranges, calibrated to each firm's gross profit and equity, resulting in a configuration that is virtually unique for each respondent. This design was developed through an iterative pretesting process with selected managers to ensure realism and relevance, and was chosen to maximize ecological validity by anchoring the investment scenario to each firm's actual financial situation while ensuring broad coverage of the experimental parameter space. A consequence of this approach is that vignette fixed effects are infeasible — including them would be equivalent to including respondent fixed effects, absorbing all between-person variation including the age effect of interest. We therefore hold project characteristics constant by including the randomized attributes directly as regression controls, which serves the same function in a virtually continuous randomization design.

We complement the vignette with external risk measures (a Holt–Laury MPL and a 7-point self-assessed risk item); full details of these measures and results are reported in the robustness section.

The sampling frame included all owner-managed Danish firms operating as limited liability companies for at least five years prior to the survey experiment. After applying exclusions for holding firms, financial institutions, conglomerate subsidiaries, and firms where

² The investment attractiveness experiment reported in this paper was embedded in a broader survey of 72 questions examining corporate investment choices and their management, partly designed to support a practical investment management tool requested by the funding organization. Only a subset of questions is used here. The complete survey is available upon request. The same data are used for a different set of research questions in Christoffersen et al. (2023), which provides descriptive evidence on managers' perceptions of risk. The present paper addresses a different research question and employs distinct empirical analyses. There is no overlap in reported tables or figures.

Table 1

Example of hypothetical investment vignette.

	<i>Worst case</i>	<i>Base case</i>	<i>Best case</i>
<i>Size of initial investment</i>	DKK 1,700,000	DKK 1,700,000	DKK 1,700,000
<i>Net loss (–) or gain (+) from the investment</i>	DKK –1,105,000	DKK 1,200,000	DKK 4,000,000
<i>Yearly return (IRR)</i>	–24%	18%	51%
<i>Time period with income from the investment</i>	2020–2028	2020–2028	2020–2028
<i>Number of years until break-even</i>	n.a.	4.7	2.2
<i>Likelihood of scenario</i>	45%	40%	15%

the top executive held concurrent leadership positions elsewhere, the sampling universe comprised 19,759 firms. Danmarks Statistik (Statistics Denmark) managed survey distribution via the CEOs' official government-linked e-Boks accounts. We received 3751 usable responses — those that could be matched to archival data on CEO and firm characteristics — corresponding to a response rate of 19.0%, which compares favorably to typical rates in top management research (e.g., Hambrick et al., 1993; Graham et al., 2015). This dataset allows us to evaluate competing behavioral mechanisms of CEO decision-making under uncertainty with high precision and strong external validity.

3.1.2. Archival data

While our key variables related to investment preferences and risk sensitivity were collected through the survey experiment, additional data—including CEO age and firm- and executive-level controls—were sourced from administrative and financial archives. Using Statistics Denmark linkages, we matched respondents to the national registry of companies and the general population register, which contains verified personal characteristics for registered executives. We extracted CEO age from this registry and constructed a binary indicator, *CEO 60 or older*, coded 1 if the CEO was aged 60 or above (0 otherwise). This threshold follows prior research that identifies meaningful cognitive-affective and motivational shifts near age 60 (e.g., Carstensen, 2006; Mather and Carstensen, 2005;

Table 2

Descriptive statistics.

	N	Mean	sd	p10	p25	p50	p75	p90
<i>Investment preference variables</i>								
Investment Attractiveness	3751	3.43	1.75	1.00	2.00	4.00	5.00	6.00
Expected Return	3751	0.12	0.12	–0.01	0.05	0.11	0.19	0.28
Standard Deviation of Returns	3751	0.25	0.11	0.13	0.18	0.24	0.31	0.40
Skewness of Returns	3751	0.13	1.24	–1.53	–0.86	0.15	1.10	1.74
Time Horizon	3751	7.50	2.68	4.00	5.00	7.00	10.00	11.00
Attitude Towards Risk (perceived)	2133	4.27	1.36	3.00	4.00	4.00	5.00	6.00
Attitude Towards Risk (Holt/Laury)	2133	4.63	2.60	0.00	3.00	5.00	6.00	7.00
<i>CEO variables</i>								
CEO Age	3751	52.93	8.95	41.00	47.00	53.00	59.00	64.00
CEO 60 or Older	3751	0.20	0.40	0.00	0.00	0.00	0.00	1.00
CEO Tenure	3751	13.12	8.55	4.00	6.00	11.00	18.00	26.00
ln(CEO Tenure)	3751	2.34	0.73	1.39	1.79	2.40	2.89	3.26
CEO Female	3751	0.11	0.31	0.00	0.00	0.00	0.00	1.00
CEO MBA	3751	0.09	0.28	0.00	0.00	0.00	0.00	0.00
CEO Net Worth***	3751	0.93	3.76	–0.92	–0.21	0.35	1.36	3.09
asinh(CEO Net Worth)	3751	0.45	1.06	–0.82	–0.20	0.34	1.12	1.85
<i>Firm variables for main tests</i>								
Firm Employees	3751	22.99	32.43	5.00	7.00	12.00	24.00	51.00
Firm Assets*	3751	23.71	45.90	2.20	4.00	8.88	22.32	55.73
Firm Age	3751	19.28	12.33	7.00	11.00	16.00	26.00	36.00
Firm ROA	3751	0.11	0.31	–0.05	0.03	0.10	0.20	0.33
ln(Firm Employees)	3751	2.66	0.86	1.61	1.95	2.48	3.18	3.93
ln(Firm Assets)	3751	2.29	1.24	0.79	1.39	2.18	3.11	4.02
ln(Firm Age)	3751	2.76	0.63	1.95	2.40	2.77	3.26	3.58
<i>Firm variables for robustness test on real outcomes</i>								
Asset Growth	3751	0.06	0.25	–0.20	–0.06	0.05	0.18	0.35
3-year Difference - Assets	3751	0.19	0.43	–0.27	–0.05	0.15	0.40	0.71
Gross Profit Growth	3751	0.03	0.28	–0.25	–0.08	0.04	0.16	0.31
3-year Difference - Gross Profit	3751	0.17	0.46	–0.30	–0.06	0.14	0.37	0.68

Notes: This table provides summary statistics for the variables used in the econometric analysis. Variables are defined in Appendix B.

* Note that the number is indicated in USD based on registered DKK data converted at exchange rate 0.15 USD/DKK.

** p10, ...,p90 refers to percentiles of the respective variables.

Sakaki et al., 2013). We report results using this threshold in the main tables and show in Fig. 2 that the main patterns are similar across a range of alternative age cutoffs.

Statistics Denmark also provides CEO-level controls: *CEO Tenure* (in years), *CEO Female* (dummy), *CEO MBA* (dummy), and *CEO Net Worth* (in DKK). These controls reflect traits known to influence executive decision-making and risk tolerance (e.g., Gomez-Mejia et al., 2019; Benischke et al., 2022). Firm-level controls (*Firm Assets*, *Firm Employees*, *Firm Return on Assets (ROA)*, *Firm Age*, and industry.) come from Orbis. Following Belenzon et al. (2019), we apply natural-log transformations to right-skewed continuous variables (*CEO Age*, *CEO Tenure*, *Firm Assets*, *Firm Employees*, and *Firm Age*) to improve normality and reduce the influence of outliers. CEO age is reported in the main tables as a binary indicator (*CEO 60 or older*); results for continuous alternatives are reported in Appendix B. Due to the right-skewed distribution of wealth (and the presence of negative values), *CEO Net Worth* is transformed using the inverse hyperbolic sine (IHS) function, a common approach in empirical wealth studies.

Appendix B provides an overview of the variable definitions, data sources, and transformations.

4. Results

4.1. Descriptive statistics

Table 2 reports summary statistics for the experimental and archival variables. The dependent variable, *Investment Attractiveness*, measured on a 7-point Likert scale, has a mean of 3.43 and a standard deviation of 1.75, indicating substantial variation in CEOs' evaluations of the investment scenarios. The *Expected Return* resulting from the probability-weighting of the worst, base and best case scenarios, has a mean of 0.12 and a standard deviation of 0.12. The mean of *Standard Deviation of Returns* is 0.25 reflecting considerable differences between potential outcomes (worst, base and best) within the vignettes. The mean *Skewness of Returns* is 0.13 suggesting

Table 3

Regressing investment attractiveness on CEO 60 or older, risk moments (standard deviation and skewness) and time horizon.

	(1)	(2)	(3)
Dependent variable:			
Investment Attractiveness			
Expected Return	0.201 (0.001)	0.195 (0.001)	0.146 (0.029)
Standard Deviation of Returns	−0.071 (0.015)	−0.071 (0.015)	−0.082 (0.012)
Skewness of Returns	0.091 (0.127)	0.095 (0.112)	0.166 (0.013)
Time Horizon	−0.032 (0.255)	−0.034 (0.222)	−0.028 (0.375)
CEO 60 or older		−0.239 (0.001)	−0.238 (0.001)
Expected Return # CEO 60 or older			0.262 (0.090)
Standard Deviation of Returns # CEO 60 or older			0.059 (0.428)
Skewness of Returns # CEO 60 or older			−0.364 (0.016)
Time Horizon # CEO 60 or older			−0.030 (0.669)
ln(CEO Tenure)	−0.08 (0.060)	−0.069 (0.107)	−0.067 (0.114)
CEO Female	−0.009 (0.920)	−0.01 (0.911)	−0.001 (0.991)
CEO Business Degree	0.133 (0.173)	0.124 (0.203)	0.124 (0.205)
asinh(CEO Net Worth)	−0.083 (0.002)	−0.066 (0.016)	−0.067 (0.015)
ln(Firm Assets)	0.164 0.000	0.165 0.000	0.163 0.000
ln(Firm Employees)	0.002 (0.961)	−0.006 (0.892)	−0.008 (0.872)
ln(Firm Age)	−0.116 (0.031)	−0.096 (0.075)	−0.096 (0.075)
Firm ROA	0.201 (0.031)	0.188 (0.036)	0.19 (0.037)
Observations	3751	3751	3751
R2	0.044	0.046	0.048

Notes: This table shows results of OLS regressions examining the relation between Investment Attractiveness and CEO 60 or older, risk moments (Standard Deviation and Skewness) and Time Horizon (as defined in Appendix B), all standardized. *p* values are presented in brackets.

that on average the vignettes are positively skewed, though the 10th and 25th percentiles are negative reflecting that not all vignettes present distributions with right tails longer than the left. Finally, *Time Horizon* has a mean of 7.5 years, (SD = 2.68) and a 10th–90th percentile spread of 7 years (from 4 to 11 years). Parameter ranges were selected to reflect managerial realism based on pilot interviews with practicing managers. This means that the tests of the suggested risk and time horizon are limited to only pertain to these limited parameter ranges, potentially decreasing the likelihood of support for the hypotheses related to the mechanisms involving these parameters. However, it also means that respondents evaluate plausible values, which may increase engagement with the vignette attributes and thus the likelihood that these attributes affect their preferences.

Turning to CEO characteristics, the mean *CEO Age* is 52.93 years, with a median of 53—consistent with prior studies (e.g., [Yim, 2013](#); [Belenzon et al., 2019](#); [O'Sullivan et al., 2021](#)). There is substantial age variation: the youngest decile (p10) is 41 or younger, and the oldest decile (p90) is 64 or older. Notably, 20% of CEOs are aged 60 or above, justifying our age threshold for testing age-related effects.

CEO Tenure is considerable, with a mean of 13 years and a median of 11, underscoring the importance of controlling for tenure when isolating the effect of CEO age. Only 11% of CEOs are female, and 9% hold a business degree (e.g., MBA), the latter being lower than comparable samples of public company executives ([Miller et al., 2015](#)). Over 25% of CEOs have a negative net worth (according to the register data used), likely due to conservative tax-based assessments of personal assets (e.g., undervalued property holdings), which may underestimate true wealth but still reflect meaningful variation.

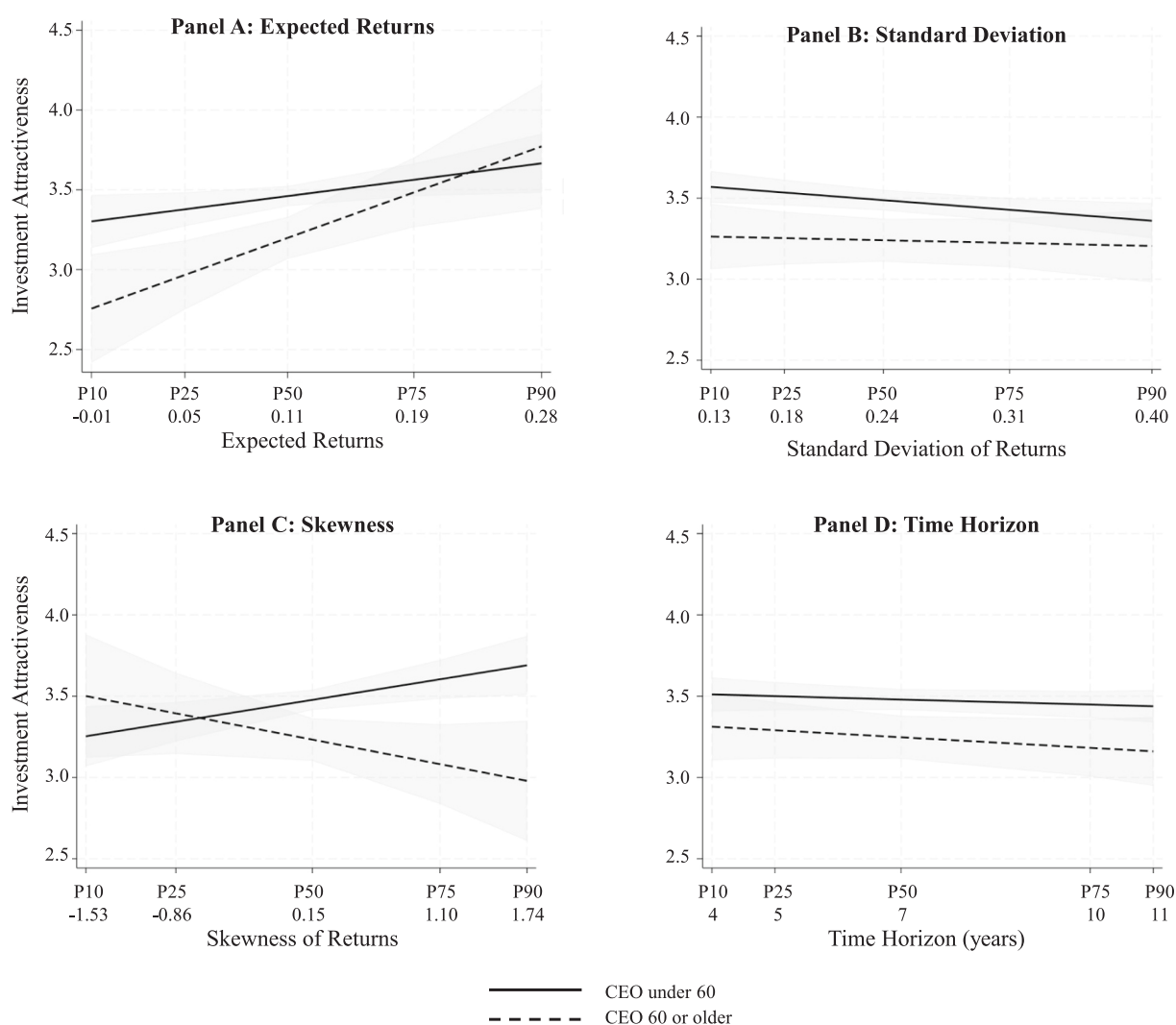


Fig. 1. Predicted investment attractiveness by project characteristics and CEO age.

Note: This figure illustrates the interaction estimates reported in [Table 3](#). Each panel plots predicted investment attractiveness for CEOs younger than 60 and CEOs aged 60 or older while varying one experimentally manipulated project characteristic over its 10th, 25th, 50th, 75th, and 90th percentiles. Predictions are obtained from the full interaction specification reported in [Table 3](#), averaging over the observed distribution of all remaining covariates. Shaded areas represent 95% confidence intervals.

Finally, the firm-level controls reveal that firm sizes are relatively small, as expected in a deliberately targeted sample of SMEs. The median for *Firm Employees* is 12 employees, while the 90th percentile size in terms of employees is 51, indicating some variation in size, which suggests it is an important control variable when assessing investment decisions. *Firm Assets* (registered in DKK millions but reported in USD converted at an exchange rate of 0.15 USD/DKK) show considerable variation. The firms are, on average, profitable with a mean (median) *Firm ROA* of 11% (10%). Finally, the *Firm Age* on average amounts to 19.3 (with a median of 16) years, showing considerable variation, which suggests that it is important to control for firm longevity as it may affect investment preferences.

4.2. Baseline responsiveness to project fundamentals

Table 3 reports OLS estimates of *Investment Attractiveness*. Column 1 includes coefficients for the first three moments of the return distribution (*Expected Return*, *Standard Deviation of Returns*, *Skewness of Returns*) as well as the *Time Horizon* along with the control variables. CEOs respond to project fundamentals in economically intuitive ways: *Expected Return* enters positively and significantly ($\beta = 0.201, p = 0.001$); *Standard Deviation of Returns* enters negatively and significantly ($\beta = -0.071, p = 0.015$); *Skewness of Returns* is positive but marginally insignificant ($\beta = 0.091, p = 0.127$). *Time Horizon* is negative and insignificant ($\beta = -0.032, p = 0.255$). These results validate the experimental design and confirm that respondents condition their evaluations on the randomized attributes.

4.3. CEO age and baseline investment attractiveness (H1)

Column 2 adds an indicator for CEO 60 or older to test H1. The coefficient for CEO 60 or older is -0.239 ($p = 0.001$), indicating that, conditional on identical returns and probabilities, CEOs aged 60+ rate the investment opportunity as less attractive than younger CEOs. This pattern is more consistent with a baseline downward shift in investment attractiveness than with broad differences in conventional risk preferences or investment horizons. The coefficient corresponds to roughly 14% of a within-sample standard deviation ($SD = 1.75$). Put differently, a project would require an expected IRR approximately 1.2 percentage points higher to offset the estimated effect of the CEO being aged 60 or older. This suggests that the estimated age effect is economically meaningful.

4.4. Horizon and risk-moment heterogeneity (H2 and H3)

Column 3 of Table 3 examines whether the negative age effect documented above reflects differences in how CEOs evaluate specific project characteristics rather than a general reduction in *Investment Attractiveness*. To test this possibility, we interact the indicator for CEO 60 or older with the four experimentally manipulated project characteristics: *Expected Return*, *Standard Deviation of Returns*, *Skewness of Returns*, and *Time Horizon*. Each variable is standardized to have mean zero and unit standard deviation prior to constructing the interaction terms, so that the main effect of CEO 60 or older in Column 3 reflects the age gap in investment attractiveness evaluated at sample-mean project characteristics. Fig. 1 provides a graphical representation of the implied prediction profiles.

The key insight from Fig. 1 is that the predicted responses to project characteristics for younger and older CEOs are quite similar. The profiles evolve in approximately parallel fashion for *Expected Return*, *Standard Deviation of Returns*, *Skewness of Returns*, and *Time Horizon*, indicating that older CEOs respond to changes in these project characteristics much like younger CEOs. At the same time, the profiles remain vertically separated over most of the observed support, consistent with the negative main effect of CEO age reported in Table 3. Thus, the figure suggests that CEO age primarily shifts the baseline attractiveness assigned to investment opportunities rather than fundamentally altering how financial project characteristics are evaluated.

The regression estimates support this interpretation. The interaction between CEO 60 or older and *Time Horizon* is negative but statistically insignificant ($\beta = -0.030, p = 0.669$), providing no support for H2. Likewise, the interaction between CEO 60 or older and *Standard Deviation of Returns* is statistically insignificant ($\beta = 0.059, p = 0.428$), providing no support for H3a. These findings indicate that older CEOs neither discount long-horizon projects more heavily nor exhibit systematically different sensitivity to return variance than younger CEOs.

The only statistically significant interaction concerns *Skewness of Returns* ($\beta = -0.364, p = 0.016$). As illustrated in Panel C of Fig. 1, younger CEOs assign increasingly higher attractiveness ratings to projects with more positively skewed return distributions, whereas this relationship is considerably weaker among CEOs aged 60 years or older. Importantly, however, this pattern is opposite to what would be expected under a conventional risk-aversion explanation. If the lower investment propensity of older CEOs primarily reflected greater aversion to downside risk, they should place relatively greater value on positively skewed projects that offer greater upside potential while holding expected return and variance constant. Instead, older CEOs appear less responsive to positive skewness.

Overall, the interaction analysis provides little evidence that CEO age systematically alters how project characteristics are evaluated. Rather, the dominant age effect appears to be an approximately parallel downward shift in investment attractiveness across the range of experimentally manipulated project characteristics. This pattern is more consistent with a baseline downward shift in investment attractiveness than with broad differences in conventional risk preferences or investment horizons.

5. Additional analyses and robustness tests

5.1. Robustness to alternative age thresholds

A potential concern is our choice of age 60 as the cutoff for classifying a CEO as “older.” Therefore, we conduct a sensitivity analysis related to this choice by rerunning the regression of column 3 in Table 3 with alternative age cut-offs ranging from 55 to 65. The

resulting coefficients for the age dummies interacted with the first three return moments (expected values, standard deviation and skewness) as well as age dummies interacted with the *Time Horizon* are plotted in Fig. 2 and show that our results are robust to alternative age cut-offs five years lower or higher than our original cut-off.

5.2. Independent risk-preference measures

Next, we explore whether independently measured risk preferences can account for the observed age–investment relationship. To address this possibility, we implemented a Holt and Laury (2002) multiple price list (MPL) task, which measures symmetric risk aversion through choices between pairs of gain-only lotteries.

In the MPL, CEOs made ten choices between a safer lottery (2000 DKK with probability p , 1600 DKK with probability $1-p$) and a riskier lottery (3850 DKK with probability p , 100 DKK with probability $1-p$), for $p \in \{0.10, 0.20, \dots, 1.00\}$. Because the MPL includes no certain option and is framed in gains, it provides a benchmark for conventional risk aversion while minimizing the role of certainty and loss framing. In addition, CEOs self-assessed their general willingness to take risks on a 7-point scale, following Dohmen et al. (2011).

The risk-preference module appeared on the final screens of a long survey, while the investment attractiveness questions underlying our main experimental results of Table 3 appeared early in the survey. In between were questions related to the firm's most recent actual investment activity, creating a natural completion barrier for less investment-active CEOs before they reached the risk-preference module. Hence the subsample of 2133 (compared to 3751 in the main analysis) may be biased towards more investment-active CEOs which should be kept in mind when interpreting the following results.

As shown in Column 1 in Table 4A, the self-reported risk-tolerance item shows no age difference. The indicator for *CEO 60 or older* is likewise unrelated to lottery-based risk aversion; the coefficient is small and statistically insignificant (Table 4A, Col. 2). Both measures correlate in expected ways with related CEO and firm traits, supporting construct validity. Taken together, and bearing in mind the subsample caveat noted above, the absence of an age gradient on two independent measures makes it unlikely that our main result is driven by generic risk aversion.

Table 4B incorporates the risk measures directly as controls in the main specification, and examines the subsample restriction that their inclusion entails. Columns (1)–(2) reproduce the main full-sample results ($N = 3751$). Columns (3)–(4) restrict to the 2133 respondents who completed the risk-preference module, without adding the risk measures. Columns (5)–(6) add the risk measures as controls on the same restricted sample.

The central finding is that the CEO 60 or older coefficient falls from -0.239 in column (2) to -0.086 in column (4) — before any risk controls are added. Adding the risk measures in columns (5)–(6) produces an essentially identical coefficient of -0.083 . The attenuation is therefore driven entirely by sample composition, not by conditioning on risk preferences: adding the risk measures changes the age coefficient by 0.003 within the subsample. This pattern holds across all controls, which also attenuate broadly when moving from the full sample to the subsample, consistent with compressed variation in investment propensity among the more active subsample respondents. Taken together, the analyses in Tables 4A and 4B indicate that the attenuation of the age effect in the subsample reflects sample composition rather than the inclusion of risk-preference controls, and that the risk measures explain little of the age–investment relationship within the subsample.

5.3. External validity: linking experimental inaction to realized investment

Finally, to assess external validity, we construct a CEO-specific *Inaction Propensity* from the experiment and relate this to real investment and activity patterns as revealed in asset growth and growth in gross profit, which is the first income statement item available for the majority of the firms in the sample due to exemptions from reporting revenue. *Inaction Propensity* is defined as the CEO-level residual from an OLS regression of *Investment Attractiveness* on the four randomized project attributes — Expected Return, Standard Deviation of Returns, Skewness of Returns, and Time Horizon — and captures each CEO's idiosyncratic tendency to rate investment opportunities above or below what the project's objective characteristics would predict. The first-stage regression includes only the randomized attributes and not CEO or firm characteristics, so the residual is orthogonal to the project attributes by construction. Since CEO and firm characteristics are not included in the first stage, the residual will in general be correlated with these variables. Spearman rank correlations (untabulated) between *Inaction Propensity* and observable CEO and firm characteristics are uniformly small in

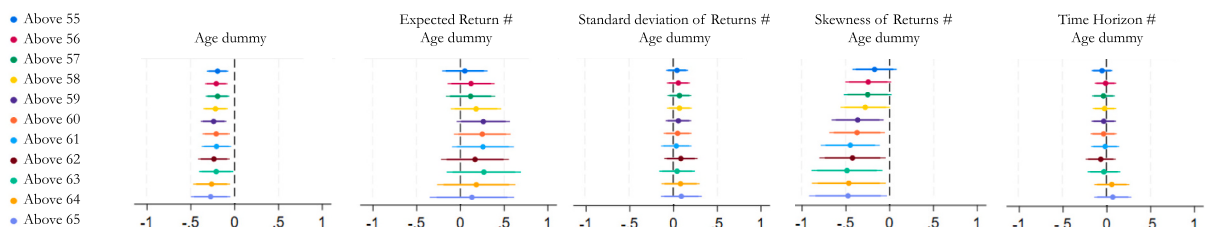


Fig. 2. Robustness of the estimated effect of CEO age on investment attractiveness.

Note: This figure summarizes the estimated effect of CEO age for alternative age cut-offs. Point estimates are shown together with 95% confidence intervals.

Table 4A
Regressing attitude towards risk on CEO 60 or older.

	(1)	(2)
	Risk attitude (self-reported)	Risk attitude (Holt-Laury)
CEO 60 or older	0.014 (0.855)	−0.198 (0.210)
ln(CEO Tenure)	−0.218 0.000	−0.072 (0.366)
CEO Female	−0.309 (0.002)	−0.473 (0.012)
CEO Business Degree	0.171 (0.037)	0.33 (0.062)
asinh(CEO Net Worth)	−0.056 (0.038)	−0.187 0.000
Observations	2133	2133
R2	0.022	0.013

Notes: This table shows results of OLS regressions examining the relation between *CEO 60 or older* and *Risk Attitudes*. *p* values are presented in brackets.

Table 4B
Comparison of age effects on investment attractiveness: full sample vs risk preference subsample.

	Full sample		Subsample answering risk attitude			
	(1)	(2)	(3)	(4)	(5)	(6)
	Dependent variable: Investment Attractiveness		Dependent variable: Investment Attractiveness			
Expected Return	0.201 (0.001)	0.195 (0.001)	0.226 (0.004)	0.225 (0.004)	0.227 (0.003)	0.226 (0.003)
Standard Deviation of Returns	−0.071 (0.015)	−0.071 (0.015)	−0.102 (0.006)	−0.103 (0.006)	−0.106 (0.004)	−0.106 (0.004)
Skewness of Returns	0.091 (0.127)	0.095 (0.112)	0.142 (0.071)	0.143 (0.070)	0.139 (0.070)	0.140 (0.070)
Time Horizon	−0.032 (0.255)	−0.034 (0.222)	−0.035 (0.348)	−0.036 (0.326)	−0.043 (0.241)	−0.045 (0.224)
CEO Age:						
CEO 60 or older		−0.239 (0.001)		−0.086 (0.394)		−0.083 (0.397)
Risk controls:						
Attitude Towards Risk (perceived)					0.21 0.000	0.21 0.000
Attitude Towards Risk (Holt/Laury)					0.042 (0.004)	0.042 (0.005)
Other controls:						
ln(CEO Tenure)	−0.08 (0.060)	−0.069 (0.107)	0.009 (0.867)	0.015 (0.784)	0.049 (0.374)	0.055 (0.323)
CEO Female	−0.009 (0.920)	−0.01 (0.911)	−0.014 (0.909)	−0.014 (0.904)	0.058 (0.622)	0.057 (0.627)
CEO Business Degree	0.133 (0.173)	0.124 (0.203)	0.005 (0.968)	0.005 (0.968)	−0.033 (0.772)	−0.033 (0.772)
asinh(CEO Net Worth)	−0.083 (0.002)	−0.066 (0.016)	−0.049 (0.155)	−0.044 (0.214)	−0.028 (0.408)	−0.023 (0.508)
ln(Firm Assets)	0.164 0.000	0.165 0.000	0.138 (0.002)	0.139 (0.002)	0.110 (0.014)	0.110 (0.013)
ln(Firm Employees)	0.002 (0.961)	−0.006 (0.892)	−0.048 (0.435)	−0.051 (0.411)	−0.048 (0.430)	−0.051 (0.405)
ln(Firm Age)	−0.116 (0.031)	−0.096 (0.075)	−0.091 (0.202)	−0.086 (0.230)	−0.064 (0.363)	−0.059 (0.403)
Firm ROA	0.201 (0.031)	0.188 (0.036)	0.202 (0.265)	0.195 (0.280)	0.229 (0.194)	0.223 (0.207)
Observations	3751	3751	2133	2133	2133	2133
R2	0.044	0.046	0.047	0.047	0.080	0.081

Notes: This table shows results of OLS regressions examining the relation between Investment Attractiveness and CEO 60 or older, return moments (Expected Return, Standard Deviation and Skewness) and Time Horizon (as defined in [Appendix B](#)) (all standardized) on different samples - without and with controls for risk attitudes. *p* values are presented in brackets.

magnitude, with the largest being 0.10 with Firm Assets. The risk-preference measures are likewise not included in the first stage and correlate modestly with Inaction Propensity (0.18 and 0.09 for self-reported and Holt–Laury measures respectively), consistent with more risk-tolerant CEOs rating investment opportunities more favorably on average. Table 5 controls for observable CEO and firm characteristics directly in the second stage, and the risk-preference measures are addressed separately in Tables 4A and 4B.

Table 5, Columns 1 (and 3) and 5 (and 7) show that *CEO 60 or older* is negatively related to investment and activity levels. This mirrors our experimental results and aligns with prior studies documenting a drop in corporate investment around the same age threshold (e.g., Belenzon et al., 2019). Columns 2 (and 4) and 6 (and 8) add the experimental *Inaction Propensity*: the CEO-level residual predicts lower realized investment and activity in administrative data. Including *Inaction Propensity* in these regressions attenuates the coefficient on *CEO 60 or older*, consistent with part of the age–investment relationship operating through the residual behavioral component captured in the experiment, beyond conventional risk preferences and investment horizons.

As an additional check on selection, we compare CEO age–outcome regressions for respondents and non-respondents by including an indicator for survey response and its interaction with the indicator CEO 60 or above. In untabulated results, the response indicator and the interaction term are statistically insignificant across all specifications (smallest p -value = 0.48), indicating no detectable differences between respondents and non-respondents either in levels of investment or in the age–investment gradient. This comparison suggests that survey participation is unlikely to drive our main inferences.

6. Discussion and conclusion

This paper studies why older CEOs evaluate and ultimately undertake fewer investments. While the dominant interpretation in corporate finance is that CEO age proxies for greater risk aversion or shorter horizons, archival investment outcomes alone make these channels hard to separate. We address this identification problem by measuring CEOs' investment decision rules directly in a large scenario-based experiment with 3751 Danish CEOs, orthogonally randomizing downside exposure, upside potential, and payout horizon.

Three main findings stand out. First, after accounting for expected returns, risk moments, and payout horizon, older CEOs evaluate otherwise identical investment opportunities as less attractive. Second, we find little evidence that older CEOs are systematically more sensitive to variance, skewness, or payout horizon. Third, an experimentally measured inaction propensity predicts lower realized investment and attenuates the age–investment relationship in administrative data.

These results clarify how CEO age should be interpreted in corporate finance. Prior work documents lower investment and fewer strategic actions among older-CEO firms, and some studies suggest this pattern is not purely selection. Our evidence suggests that the age gradient cannot readily be explained by conventional risk preferences or investment horizons alone. Instead, a substantial residual difference in investment evaluation remains. We interpret this residual as reflecting a broader tendency towards lower investment propensity, while recognizing that several underlying mechanisms may contribute to this pattern. Put differently, CEO age appears to

Table 5
Regressing investments on CEO 60 or older and inaction propensity (residuals from experimental regressions).

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Asset Growth	Asset Growth	3-year Difference - Assets	3-year Difference - Assets	Gross Profit Growth	Gross Profit Growth	3-year Difference - Gross Profit	3-year Difference - Gross Profit
CEO 60 or older	−0.029 0.000	−0.027 0.000	−0.128 0.000	−0.125 0.000	−0.034 0.000	−0.032 0.000	−0.151 0.000	−0.148 0.000
ln(Firm Assets)	0.012 0.000	0.011 0.000	0.063 0.000	0.061 0.000	0.013 0.000	0.013 0.000	0.035 0.000	0.033 0.000
ln(Firm Age)	−0.039 0.000	−0.039 0.000			−0.036 0.000	−0.036 0.000		
ln(CEO Tenure)	−0.010 0.000	−0.010 0.000			−0.015 0.000	−0.015 0.000		
CEO Female	−0.014 (0.016)	−0.014 (0.020)			−0.025 0.000	−0.024 0.000		
CEO Business Degree	−0.005 (0.406)	−0.005 (0.395)			0.003 (0.730)	0.002 (0.741)		
asinh(CEO Net Worth)	−0.006 (0.001)	−0.005 (0.002)			−0.005 (0.016)	−0.004 (0.022)		
Inaction Propensity		0.004 0.000	0.010 0.000			0.004 (0.001)		0.012 0.000
Constant	0.149 0.000	0.150 0.000	0.066 0.000	0.069 0.000	0.153 0.000	0.154 0.000	0.111 0.000	0.114 0.000
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	21,156	21,156	14,571	14,571	20,933	20,933	14,353	14,353
R ²	0.131	0.135	0.043	0.044	0.111	0.114	0.022	0.024

Notes: This table shows results of examining the relation between investment behavior and CEO 60 or older as well as Inaction Propensity which is residuals from regressing Investment Attractiveness on the experimental variables of column 3 in Table 3 (i.e. Expected Return, Standard Deviation of Returns, Skewness of Returns).

map onto a residual managerial-style dimension influencing investment evaluations beyond conventional risk preferences and investment horizons. The findings also help reconcile an empirical puzzle that is difficult to explain using conventional risk-aversion accounts alone. Older leaders frequently exhibit strategic inertia—not only initiating fewer investments but also undertaking fewer corrective actions such as restructuring or divestment. This joint pattern is consistent with a broad class of mechanisms that reduce investment propensity and thereby make managers less likely to initiate strategic action, although our evidence does not distinguish among these underlying mechanisms.

One possible mechanism concerns age-related shifts in affective processing: the negativity bias weakens and a positivity effect emerges in later life (Taylor, 1991; Baumeister et al., 2001; Rozin and Royzman, 2001; Mather and Carstensen, 2005; Carstensen and Mikels, 2005). In decisions with an explicit sure default, such shifts can reinforce default attraction by increasing the salience of avoiding disappointing outcomes or ex post blame. We view this as one plausible mechanism among several that may contribute to lower investment propensity with age.

For corporate finance research on CEO effects, the paper offers three contributions. First, it provides direct evidence on how CEO age shapes evaluation of investment opportunities, complementing archival CEO-effects approaches that infer preferences from realized policies (Bertrand and Schoar, 2003; Malmendier and Tate, 2008; Graham et al., 2013; Bennedsen et al., 2020). Second, it shows that the age–investment gradient is not readily explained by conventional risk preferences or investment horizons alone. Instead, we document a substantial residual age-related difference in investment evaluation that is consistent with a broader residual tendency towards lower investment propensity beyond conventional risk preferences. Third, it links experimentally measured decision components to realized corporate investment: a CEO-specific inaction propensity extracted from randomized evaluations predicts lower field investment and reduces the reduced-form age–investment association.

The findings have implications for boards and capital-allocation processes, although the appropriate governance response depends on the mechanisms underlying the residual age–investment relationship. If part of this residual reflects implementation frictions, greater default attraction, or other influences that reduce investment propensity beyond conventional risk preferences and investment horizons, boards may benefit from lowering unnecessary barriers to initiating valuable investments. Practical steps could include staged commitments and real-options sequencing, delegating early screening and due diligence to independent committees, instituting pre-specified decision protocols that reduce cognitive and coordination burdens, and structured reviews that make the counterfactual cost of inaction explicit (e.g., “what is the cost of not investing?”). At the same time, reducing barriers to action may increase the risk of approving marginal or ill-conceived projects; boards should therefore combine such initiatives with stronger early screening, staged commitments, and explicit ex ante performance gates. As CEO tenures lengthen, age-diverse top management teams and formalized challenge roles may also help sustain adaptive capital allocation by ensuring that alternative investment opportunities receive systematic consideration.

This study has limitations that suggest directions for future work. Our sample comprises owner-managed Danish firms, where CEOs typically have concentrated authority and socioemotional stakes; it remains important to examine whether the broader residual tendency towards lower investment propensity documented here generalizes to large public firms with layered decision processes and different incentive structures. Future research could disentangle the relative importance of the different mechanisms that may contribute to lower investment propensity and examine when each is most influential—for example, in more complex firms, industries with greater technological uncertainty, or governance regimes that raise personal accountability costs. Extending the experimental design to loss-domain choices (shutdowns, layoffs, exits) would also help determine whether certainty weighting and other behavioral mechanisms generate age-related differences in break-even or loss-realization behavior.

In sum, our evidence suggests that the age–investment gradient is not readily explained by conventional risk preferences or investment horizons. Instead, we document a substantial residual age-related difference in investment evaluation that predicts realized corporate investment and is consistent with a broader tendency towards lower investment propensity arising through multiple possible mechanisms.

CRedit authorship contribution statement

Steffen Brenner: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jeppe Christoffersen:** Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Project administration, Investigation, Funding acquisition, Formal analysis, Data curation. **Torben Andersen:** Writing – review & editing, Investigation, Formal analysis, Conceptualization. **Thomas Plenborg:** Writing – review & editing, Validation, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interest or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Parametrization and example

Experimental participants were presented with a hypothetical business opportunity, varying in several key attributes that may determine how managers assess the risk and attractiveness of the business opportunity. The managers assessed an investment project with three scenarios (worst-case, base-case, and best-case), for which they were informed about the size of the initial investment and the time-period with income from the investment. For each of the scenarios, participants were informed about four key performance indicators: (i) the likelihood of the scenario, (ii) the net loss or gain from the investment, (iii) the yearly internal rate of return (IRR), and (iv) the number of years to break even (B/E). The participant received information about the variables but with randomly drawn values (within a pre-specified range) as researcher-controlled variation in explanatory variables. Importantly, the estimates of the key characteristics were scaled by a firm size factor – based on the company's gross profits and its total equity, obtained from the company's financial statements – to ensure that the investment project represents reasonable stakes across all respondents. The estimated values were chosen so they could strike a satisfactory balance between wide variation in the variables and managers still perceiving the values as being within the range relevant for their managerial decision-making. These values were based on an iterative pilot approach where we tested different ranges of the parameters with the participating managers. [Table A1](#) provides an example of variables shown to a participant and [Table A2](#) summarizes the parametrization.

Table A1

Example of hypothetical investment opportunity shown.

	Worst case	Base case	Best case
Size of initial investment	DKK 1,700,000	DKK 1,700,000	DKK 1,700,000
Net loss (–) or gain (+) from the investment	DKK –1,105,000	DKK 1,200,000	DKK 4,000,000
Yearly return (IRR)	–24%	18%	51%
Time period with income from the investment	2020–2028	2020–2028	2020–2028
Number of years until break-even	n.a.	4.7	2.2
Likelihood of scenario	45%	40%	15%

Table A2

Parametrization of the investment project in three scenarios.

	Worst case ($i = 1$)	Base case ($i = 2$)	Best case ($i = 3$)
Investment size (I)	$I = i \cdot f$ with $i \in \{1\%, 2\%, \dots, 100\%\}$ and $f = 0.25 \cdot \text{gross return} + 0.50 \cdot \text{total equity}$		
Time A ($s_A - e_A$)	From $s_A \in \{2020, 2021\}$ until $e_A [=s_A + t_A]$ with $t_A \in \{2, 3, \dots, 10\}$		
Net loss/gain (π_i)	$\pi_1 = I \cdot r_{1A}$ with $r_{1A} \in \{-0.95, -0.90, \dots, -0.25\}$	$\pi_2 = I \cdot r_{2A} \cdot (e_A - 2019)$ with $r_{2A} \in \{0.010, 0.011, \dots, 0.100\}$	$\pi_3 = I \cdot r_{3A} \cdot (e_A - 2019)$ with $r_{3A} \in \{0.11, 0.12, \dots, 0.75\}$
Rate of return (irr_i)	$irr_1 \in \{-83\%, -82\%, \dots, -4\%\}$ (determined by r_{1A} , s_A and t_A)	$irr_2 \in \{3\%, 4\%, \dots, 20\%\}$ (determined by r_{2A} , s_A and t_A)	$irr_3 \in \{15\%, 16\%, \dots, 133\%\}$ (determined by r_{3A} , s_A and t_A)
Time B ($s_B - e_B$)	From $s_B \in \{2020, 2021\}$ until $e_B [=s_B + t_B]$ with $t_B \in \{2, 3, \dots, 10\}$		
Break-even (b_i)	$b_1 = \text{"n.a."}$	$b_2 \in \{1.5, 1.6, \dots, 9.1\}$ (determined by r_{2B} , s_B and t_B)	$b_3 \in \{0.6, 0.7, \dots, 5.4\}$ (determined by r_{3B} , s_B and t_B)
Likelihood (p_i)	$p_1 \in \{5\%, 10\%, \dots, 55\%\}$ if $p_2 = 40\%$ $p_1 \in \{5\%, 10\%, \dots, 45\%\}$ if $p_2 > 50\%$ $p_1 \in \{5\%, 10\%, \dots, 40\%\}$ if $p_2 > 55\%$ etc.	$p_2 \in \{40\%, 45\%, \dots, 70\%\}$	$1 - p_2 - p_1$

The attributes (i , s_A , t_A , r_A , s_B , t_B , r_B , p_i) – as defined in the table – were randomly drawn from uniform distributions to determine the parameters of the business opportunity. The scenarios were scaled by a factor (f), based on the gross return and total equity obtained from the company's accounting data.

Explanatory example

We demonstrate our procedure for arriving at the different values in the [Table A2](#) with an example of a firm with gross profit of DKK m 6.186 and equity of DKKm 4.461. As we want to present numbers within a plausible range for each of the firms, we use accounting information to determine a size factor (f) which we let depend on Gross profit (multiplied with 0.25) as well as Equity (multiplied with 0.5). In this case the size factor is DKKm 3.777 ($0.25 \cdot \text{DKK m 6.186} + 0.50 \cdot \text{DKK m 4.461}$).

Size of initial investment

To estimate investment size, we multiply the size factor by 46% (s) which is drawn randomly from $i \in \{1\%, 2\%, \dots, 100\%\}$. The resulting investment size is DKK m 1.737420 which we round to DKK m 1.7 (I).

Net loss (–) or gain (+) from the investment

The next item seen by the participant is Net loss / gain (π_i), which is a loss in the Worst-Case scenario and a gain in the Base Case and Best Case scenarios. In the Base Case and Best-Case scenarios this performance depends on the Time A ($s_A - e_A$). In this example, the cash

inflows start in 2020 (drawn randomly from $s_A \in \{2020, 2021\}$ and end in 2026 (e_A) arrived at by adding 6 years (t_A) drawn randomly from $t_A \in \{2, 3, \dots, 10\}$. We treat cash flows as if they materialize mid-year. For the Base Case, we multiply the investment size I with 10% (r_{2A}) drawn randomly from $r_{2A} \in \{0.010, 0.011, \dots, 0.100\}$ and then with the number of years from the investment is made (in 2019 for all firms) till it ends ($e_A - 2019$) to arrive at a Base Case *Net Gain* of DKK m 1.19 [=1.7*10%*(2026–2019)] which we round to DKK m 1.2. Similarly for the Best Case, we multiply the investment size (I) with 34% (r_{3A}) drawn randomly from $r_{3A} \in \{0.11, 0.12, \dots, 0.75\}$ and then with the number of years from the investment is taken till it ends ($e_A - 2019$) to arrive at a Best Case *Net Gain* of DKK m 4.046 [=1.7*34%*(2026–2019)] which we round to DKK m 4.0. For the Worst-Case *Net Loss* we multiply the investment size I with –65% (r_{1A}) drawn randomly from $r_{1A} \in \{-0.95, -0.90, \dots, -0.25\}$ and then with the number of years from the investment is taken till it ends ($e_A - 2019$) to arrive at a Worst Case *Net Loss* of DKK m –1.105 rounded to DKK m –1.1.

Yearly return (IRR)

To arrive at the *Rate of return* (irr_i), we assume that a negative cash flow ($-I$) falls in 2019, whereas the positive cash flows ($I + \pi_i$) for the three scenarios (*Worst Case* ($i = 1$), *Base Case* ($i = 2$), *Best Case* ($i = 3$)) are evenly distributed over the specified time period. Hence, the Base Case cash flows (in DKK m) used to perform the IRR calculation are as follows:

$$2019 : -1.7, 2020-2023 : (1.7 + 1.2)/6 = 0.483333$$

using the formula

$$2019 : -I, 2020-2023 : (I + \pi_2)/t_A$$

We then calculate the IRR of this cash flow stream to reach at the result of $IRR_2 = 18\%$.

Using the same approach for the Worst Case, we arrive at the following cash flow (in DKK m) for the calculation of IRR:

$$2019 : -1.7, 2020-2023 : (1.7 - 1.105)/6 = 0.99167$$

using the formula

$$2019 : -I, 2020-2023 : (I + \pi_1)/t_A$$

We then calculate the IRR of this cash flow stream resulting in an $IRR_1 = -24\%$.

Finally, for the Best Case we use the following cash flows (in DKK m) to calculate IRR:

$$2019 : -1.7, 2020-2023 : (1.7 + 4)/6 = 0.950$$

using the formula

$$2019 : -I, 2020-2023 : (I + \pi_3)/t_A$$

We then calculate the IRR of this cash flow stream with result $IRR_3 = 51\%$.

As noted in the table, IRR is determined solely by r_{1A} (r_{2A} and r_{3A}), s_A and t_A and is thus uncorrelated with the investment size I .

Time period with income from the investment

We simulate a new *Time B* ($s_B - e_B$). In this example we draw $s_B = 2020$ and $t_B = 8$ resulting in $e_B = 2028$ and therefore presents the time-period 2020–2028 to the participant.

Time B differs from Time A which means that it is uncorrelated with the absolute and relative returns (*Net loss* (–) or *gain* (+) and *Yearly return* (IRR)).

Number of years until break-even

The number of years in the Base Case and Best Case until Breakeven are based on a simulation of cash flows generated by the same process to generate *Net loss* (–) or *gain* (+) from the investment but with other values for the different parameters. In this example we randomly draw $r_{2B} = 8\%$ and $r_{3B} = 29\%$.

This determines *Break-even* (b_i) that is correlated with the time-period seen by the participants but uncorrelated with the absolute and relative returns (*Net loss* (–) or *gain* (+) and *Yearly return* (IRR)).

The specific calculation of *Break-even* (b_i) illustrated for the Base Case, depends on the time from the investment is undertaken (2019) till cash inflows start (s_B) and the ratio between the investment I and the yearly cash flows $(I + I * r_{2B} * (e_B - 2019))/t_B$ stipulated by the following formula $[s_A - 2019 + I / ((I + I * r_{2B} * (e_B - 2019)) / t_B) - 1]$ which can be rewritten independent on investment size I as $[s_A - 2019 + 1 / ((1 + 1 * r_{2B} * (e_B - 2019)) / t_B) - 1]$. With the specific parameters drawn in our example this results in Base Case *Years till Breakeven* of 4.7 [=2020 – 2019 + 1 / ((1 + 1 * 8% * (2028 – 2019)) / 8) – 1]. Using the same approach but with $r_{3B} = 29\%$, we get Best Case *Years till Breakeven* of 2.2 [=2020 – 2019 + 1 / ((1 + 1 * 8% * (2028 – 2019)) / 8) – 1].

Likelihood of scenario

The final parameter estimated was the *Likelihood* (p_i) of the three scenarios ($i: 1, 2, 3$). First, we estimate the likelihood of the Base Case scenario p_2 with likelihoods drawn from $p_2 \in \{40\%, 45\%, \dots, 70\%\}$. The remaining $1 - p_2$ are distributed between the Worst Case and Best-Case likelihoods p_1 and p_3 randomly in chunks of 5 percentage points drawn from $p_1 \in \{5\%, 10\%, \dots, 55\%\}$ and $p_3 = 1 - p_2 - p_1$ which results in p_1 and p_3 with same underlying distributions. In this example, we reach at $p_1 = 40\%$, $p_2 = 45\%$, and $p_3 = 15\%$.

Appendix B. Variable definitions – variables for experimental analysis

Variable	Definition	Source	Transformations
<i>Investment preference variables:</i>			
Investment Attractiveness	Likert scale response (1–7) to question about attractiveness of a hypothetical investment (see example of vignette below)	Survey Experiment	
Expected Return	$p_1 * irr_1 + p_2 * irr_2 + p_3 * irr_3$		Winsorization at 1st and 99th
Standard Deviation of Returns	$((p_1 * (irr_1 - Exp. Ret.)_2 + p_2 * (irr_2 - Exp. Ret.)_2 + p_3 * (irr_3 - Exp. Ret.)_2)_{0.5}$		Winsorization at 1st and 99th
Skewness of Returns	$3 * (irr_2 - Exp. Ret.)_2 / Std. Dev. of Returns$		Winsorization at 1st and 99th
Time Horizon	$Time_B - Time_A$		
Attitude Towards Risk (perceived)	Likert scale response (1–7) to question about personal willingness to take risk (Dohmen et al., 2011)	Survey Experiment	
Attitude Towards Risk (Holt/Laury)	Experimentally elicited risk preference (Holt and Laury, 2002)	Survey Experiment	
<i>CEO variables:</i>			
CEO Age	Age as of 1. of January (based on the variable ALDER in the general population registry (BEF))	Statistics Denmark Registry Data	Log and winsorization at 1st and 99th
CEO Tenure	Year of survey response (2019) - year of first employment in firm	Statistics Denmark Registry Data	Log and winsorization at 1st and 99th
CEO Female	Dummy variable (1/0) taking on the value 1 if the individual is registered as female (based on the variable KOEN in the general population registry (BEF))	Statistics Denmark Registry Data	
CEO Business Degree	Dummy variable (1/0) taking on the value 1 if the individual has a master's degree in business or economics dominated domains (based on the variable HFAUDD of the educational registry UDDA)	Statistics Denmark Registry Data	
CEO Net Worth	The difference between the tax values of assets (incl. property and savings) and liabilities (incl. bank and mortgage debt) (based on the variables QAKTIVF_NY05-QPASSIVN in the tax registry (IND))	Statistics Denmark Registry Data	Inverse Hyperbolic Function and winsorization at 1st and 99th
<i>Firm variables:</i>			
Firm Assets	Total assets, 2019	Orbis	Log and winsorization at 1st and 99th
Firm Employees	Number of employees, 2019	Orbis	Log and winsorization at 1st and 99th
Firm ROA	Operating Profit, 2019 / Total assets, 2019	Orbis	Log and winsorization at 1st and 99th
Firm Age	Year of survey response (2019) – year of incorporation	Orbis	Log and winsorization at 1st and 99th

Q1: Please look at the information below and consider how attractive the investment – which is to take place this year – is to your company in its current economic situation.

	<i>Worst Case</i>	<i>Base Case</i>	<i>Best Case</i>
Size of initial investment	DKK 1,700,000	DKK 1,700,000	DKK 1,700,000
Net loss (-) or gain (+) from the investment	DKK -1,105,000	DKK 1,200,000	DKK 4,000,000
Yearly return (IRR)	-24%	8%	51%
Time period with income from the investment	2020–2028	2020–2028	2020–2028
Number of years until break-even	n.a.	4.7	2.2
Likelihood of scenario	45%	40%	15%

Parameters varied between companies as specified in Appendix A

Q2: Please tick all those scenarios for which you made estimates ahead of your most recent decision to pursue or not pursue a significant business-related investment.

- ☐ Worst case – considerably worse than base case
☐ Base case

- ☐ Best case – considerably better than base case
☐ Other scenarios beyond the three mentioned
☐ Not relevant

Appendix C. Variable definitions – variables for archival analysis

Variable	Definition	Source	Transformations
Investments and outcomes			
Asset growth	Total assets _{t-1} / Total assets _t	Orbis	Log and winsorization at 1st and 99th
Gross profit growth	Gross profit _{t-1} / Gross profit _t	Orbis	Log and winsorization at 1st and 99th
CEO variables			
CEO Age	Age as of 1. of January (based on the variable ALDER in the general population registry (BEF))	Statistics Denmark Registry Data	Log and winsorization at 1st and 99th
CEO Tenure	Year of survey response (2019) – year of first employment in firm	Statistics Denmark Registry Data	Log and winsorization at 1st and 99th
CEO Female	Dummy variable (1/0) taking on the value 1 if the individual is registered as female (based on the variable KOEN in the general population registry (BEF))	Statistics Denmark Registry Data	
CEO Business Degree	Dummy variable (1/0) taking on the value 1 if the individual has a master's degree in business or economics dominated domains (based on the variable HFAUDD of the educational registry UDDA)	Statistics Denmark Registry Data	
CEO Net Worth	The difference between the tax values of assets (incl. property and savings) and liabilities (incl. bank and mortgage debt) (based on the variables QAKTIVF_NY05-QPASSIVN in the tax registry (IND))	Statistics Denmark Registry Data	Inverse Hyperbolic Function and winsorization at 1st and 99th
Firm variables			
Firm Assets	Total assets, 2019	Orbis	Log and winsorization at 1st and 99th
Firm Age	Year of survey response (2019) – year of incorporation	Orbis	Log and winsorization at 1st and 99th

Data availability

The authors do not have permission to share data.

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